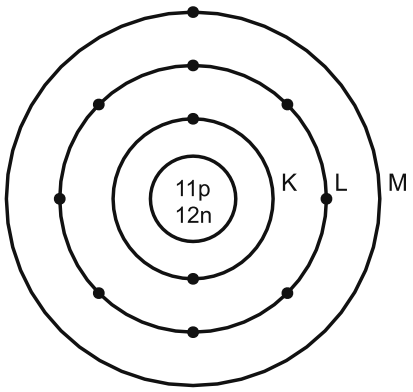
Model Answer and Marking Scheme - Chemistry IA 1

Total: 50 marks (10 x 5). Mark splits are given in [brackets]. Accept equivalent correct wording and methods.

Q1. Atomic structure [5 marks]

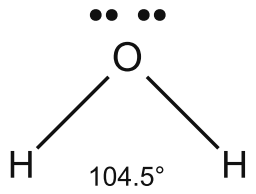


Sodium (Na)

Nucleus contains 11 protons and 12 neutrons (A - Z = 23 - 11). Electrons: K shell 2, L shell 8, M shell 1. Electronic configuration: 2, 8, 1 (1s2 2s2 2p6 3s1). Valency = 1: sodium loses its single outer electron to form Na+.

Marking: Nucleus 11p, 12n [1]; shells drawn K2 L8 M1 [2]; configuration 2,8,1 [1]; valency 1 with reason [1].

Q2. Shape of the water molecule [5 marks]



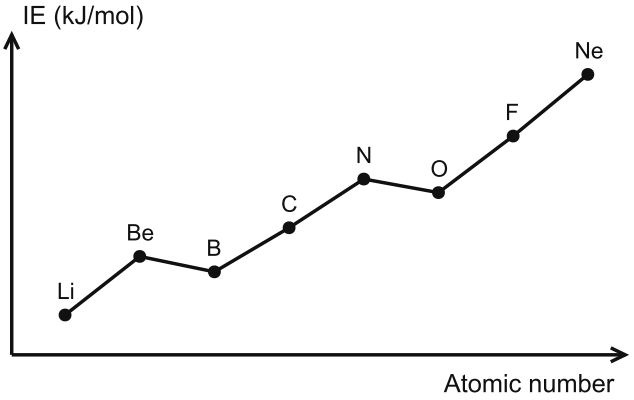
bent (V-shaped)

Oxygen forms two O-H bond pairs and keeps two lone pairs. The four electron pairs are arranged tetrahedrally, but the molecule is bent (V-shaped).

Bond angle = 104.5°. Lone pair-lone pair repulsion is greater than bond pair-bond pair repulsion, so the lone pairs push the O-H bonds closer together, reducing the angle below 109.5°.

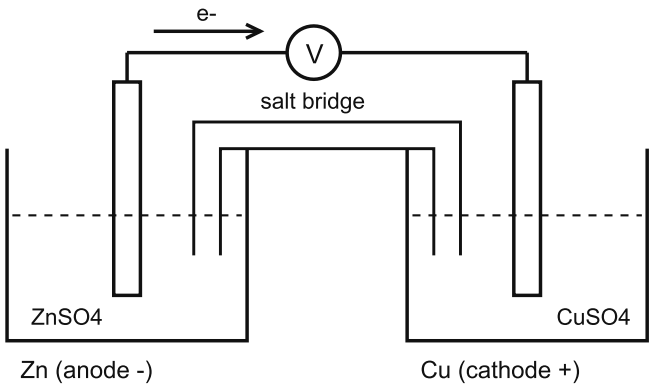
Marking: Lewis structure with 2 bond pairs and 2 lone pairs on O [2]; bent shape drawn [1]; angle 104.5° [1]; lone-pair repulsion explanation [1].

Q3. Ionisation energy [5 marks]



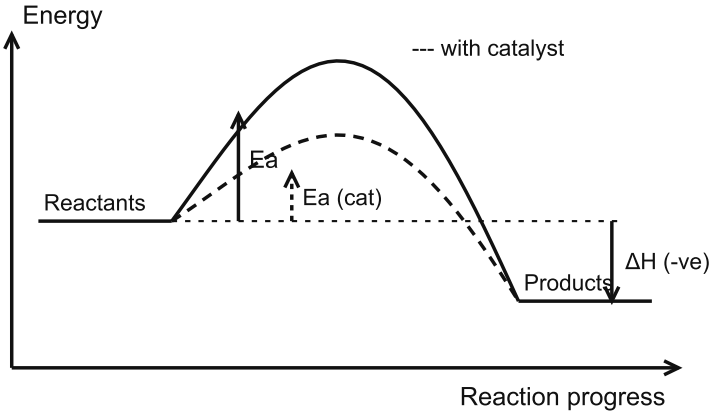
Graph rises from Li to Ne with two small drops, at B and at O.
General trend: across the period the nuclear charge increases while electrons are added to the same shell, so atomic radius decreases and the outer electron is held more strongly; ionisation energy increases.
Drop at B: the electron removed is a 2p electron, which is higher in energy and shielded by the 2s electrons.
Drop at O: the electron removed is from a doubly occupied 2p orbital; repulsion between the paired electrons makes it easier to remove.
Marking: Graph increasing with dips at B and O [2]; trend explained by nuclear charge / radius [1]; B explained (2p, shielding) [1]; O explained (paired-electron repulsion) [1].

Q4. Daniell cell [5 marks]



Zinc rod in ZnSO4 solution, copper rod in CuSO4 solution, joined by a salt bridge (KCl/KNO3); external wire with voltmeter. Electrons flow from Zn to Cu in the external circuit.
Anode (oxidation): Zn -> Zn2+ + 2e-
Cathode (reduction): Cu2+ + 2e- -> Cu
E°cell = E°cathode - E°anode = 0.34 - (-0.76) = +1.10 V
Marking: Labelled diagram with both half-cells, salt bridge and voltmeter, electron flow Zn to Cu [2]; anode reaction [1]; cathode reaction [1]; EMF 1.10 V [1].

Q5. Catalysis [5 marks]

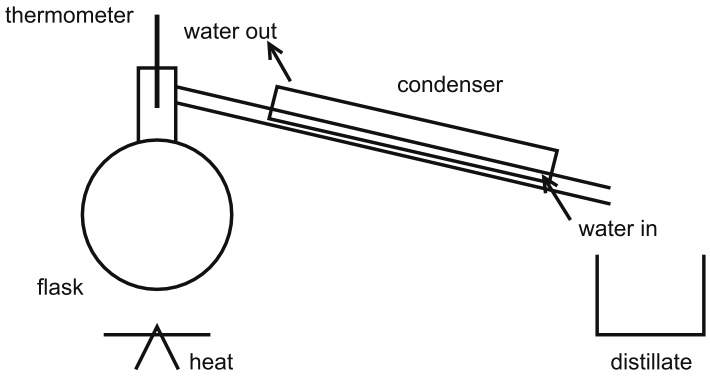


For an exothermic reaction the products lie below the reactants (ΔH negative). The catalysed pathway (dashed) has a lower peak, so Ea (cat) < Ea, while ΔH is unchanged.

A catalyst provides an alternative reaction pathway with lower activation energy, so a larger fraction of colliding molecules have energy equal to or greater than Ea, and the rate increases. The catalyst is not used up.

Marking: Exothermic profile, products lower [1]; catalysed path with lower peak [1]; Ea, Ea(cat) and ΔH labelled [1.5]; explanation: alternative path, lower Ea, more effective collisions [1.5].

Q6. Simple distillation [5 marks]



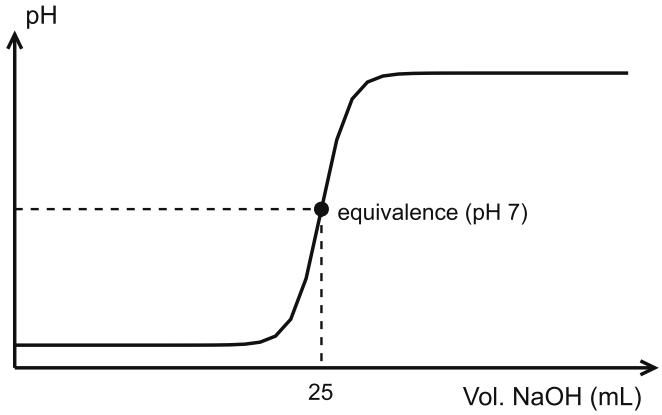
Apparatus: round-bottom flask heated from below, thermometer bulb at the side arm, Liebig condenser sloping down to a receiving flask.

The condenser cools the hot vapour so that it condenses back to liquid (the distillate).

Cooling water enters at the bottom and leaves at the top so the condenser jacket stays completely full of water and the coldest water meets the vapour just before it leaves (counter-current), giving efficient cooling.

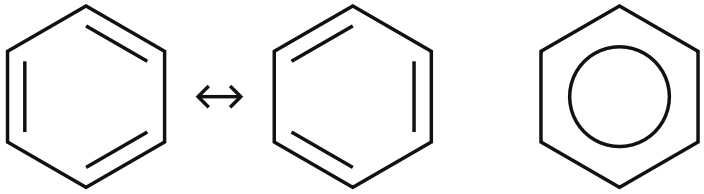
Marking: Correct apparatus drawn (flask, heat, thermometer at side arm, condenser, receiver) [2]; labels [1]; purpose of condenser [1]; water in at bottom with reason [1].

Q7. Acid-base titration [5 marks]



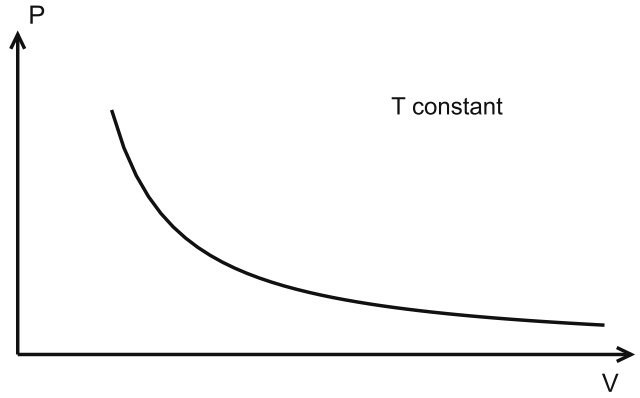
The curve starts at about pH 1, rises slowly, then rises very steeply from about pH 3 to pH 11 near the equivalence point, then levels off near pH 13.
Equivalence point: pH 7 (strong acid + strong base).
moles HCl = 0.10 × 0.0250 = 0.00250 mol, so V(NaOH) = 0.00250 / 0.10 = 0.0250 L = 25.0 mL.
Indicator: phenolphthalein (methyl orange is also acceptable).
Marking: S-shaped curve with steep rise [2]; equivalence point at pH 7 marked [1]; 25.0 mL calculated [1]; suitable indicator [1].

Q8. Benzene [5 marks]



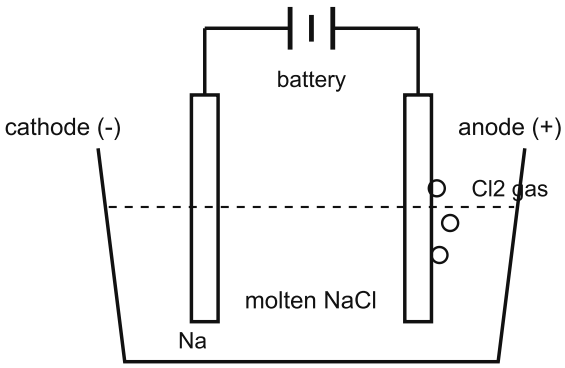
Two Kekulé structures with alternating double bonds, linked by a double-headed resonance arrow; the real molecule is the resonance hybrid (hexagon with a circle) in which all six C-C bonds are equal (139 pm). The six π electrons are delocalised over the ring, which gives benzene extra stability (resonance / delocalisation energy). Addition would destroy this delocalised (aromatic) system, whereas substitution keeps it, so benzene prefers substitution.
Marking: Two Kekulé structures with resonance arrow [2]; resonance hybrid with circle [1]; delocalised π electrons give extra stability [1]; addition destroys aromaticity, substitution retains it [1].

Q9. Boyle's law [5 marks]



Boyle's law: at constant temperature, the pressure of a fixed mass of gas is inversely proportional to its volume (PV = constant).
The P-V graph is a hyperbola (curve), since P ∝ 1/V.
P1V1 = P2V2: 2.0 × 3.0 = 6.0 × V2, so V2 = 1.0 L.
Marking: Correct statement [1]; hyperbolic P-V curve with labelled axes [2]; P1V1 = P2V2 used [1]; V2 = 1.0 L [1].

Q10. Electrolysis [5 marks]



Molten NaCl in a crucible with two inert (graphite) electrodes connected to a battery. Na+ ions move to the cathode (-) and Cl- ions move to the anode (+).
Cathode (reduction): Na+ + e- -> Na
Anode (oxidation): 2Cl- -> Cl2 + 2e-
Products: sodium metal at the cathode, chlorine gas at the anode.
Marking: Labelled diagram with molten NaCl, electrodes, battery, cathode (-) and anode (+) [2]; cathode reaction [1]; anode reaction [1]; products named [1].